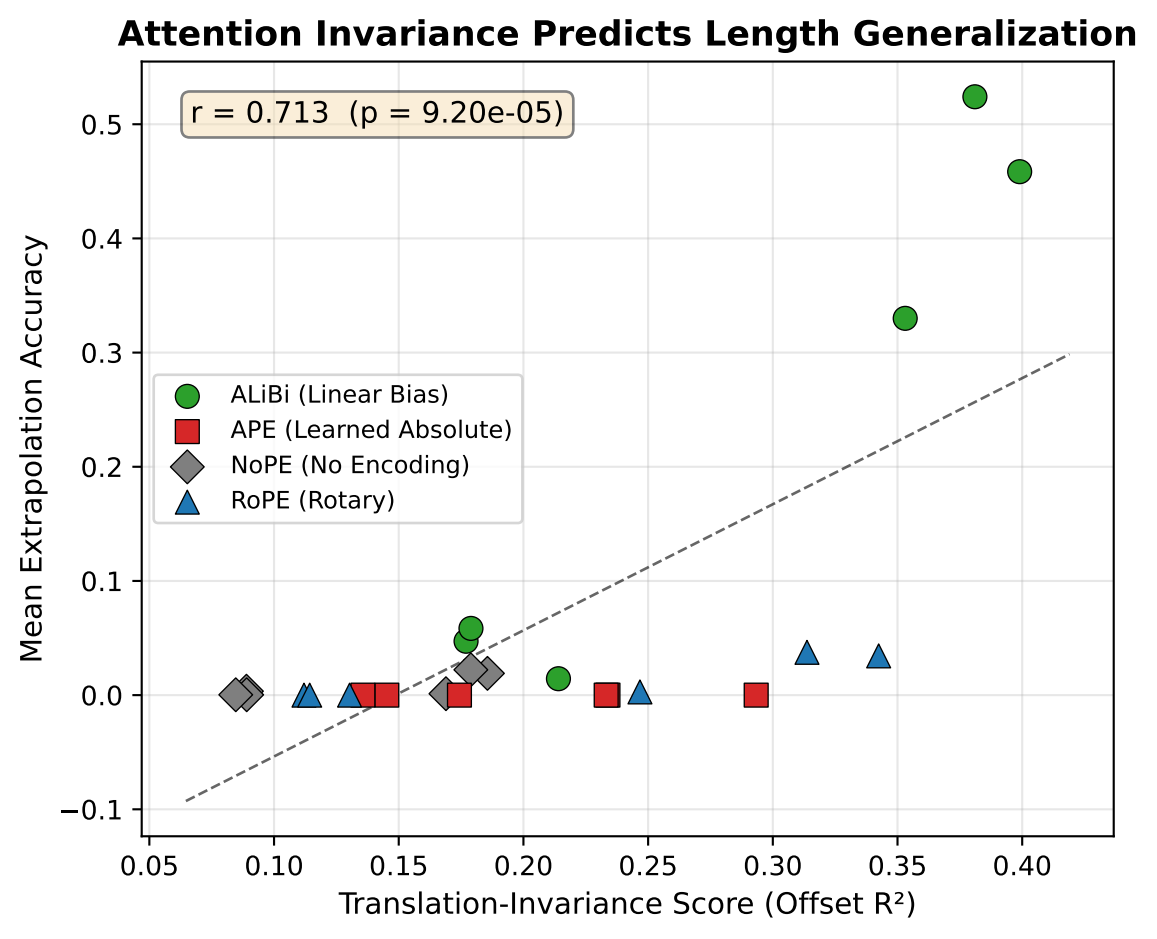
Attention Invariance Predicts Length Generalization
r = 0.713 (p = 9.20e-05)
ALiBi (Linear Bias)
APE (Learned Absolute)
NoPE (No Encoding)
RoPE (Rotary)
Mean Extrapolation Accuracy
Translation-Invariance Score (Offset R²)
0.5
0.4
0.3
0.2
0.1
0.0
−0.1
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40